Definition

Let $f = \langle \mathsf{st}, \mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{D} \rangle$ be a LTI system from $l \in \mathbb{N}$ to $m \in \mathbb{N}$. Given factorizations $l = i + k$ and $m = j + k$, we define the LTI system $\mathrm{Tr}^{k}_{i,j}\, f$ as

$$\langle \mathsf{st}, \mathbf{A} + \mathbf{B}^{[:,i+1:i+k]}\mathbf{E}\mathbf{C}^{[j+1:j+k,:]}, \mathbf{C}^{[1:j,:]} + \mathbf{D}^{[1:j,i+1:i+k]}\mathbf{E}\mathbf{C}^{[j+1:j+k,:]}, \\ \mathbf{D}^{[1:j,1:i]} + \mathbf{D}^{[1:j,i+1:i+k]}\mathbf{E}\mathbf{D}^{[j+1:j+k,1:i]} \rangle,$$

where $\mathbf{E} = (\mathbf{1} - \mathbf{D}^{[j+1:j+k,i+1:i+k]})^{-1}$ and $\mathbf{D}$ is chosen to make the expression invertible.